Activity 1.4 – 32 × 32 Register File

1. Design Approach:

Designed a 32 × 32 register file using Verilog HDL. It contains 32 registers, each having 32 bits, with two read ports and one write port. Register x0 always returns zero, and same-cycle write bypassing is included.

2. Verification Steps:

- Created a separate testbench.
- Tested register write and read operations.
- Verified both read ports.
- Checked that register x0 always returns zero.
- Tested same-cycle write bypassing.
- Applied 200 pseudo-random read/write transactions.
- Compared outputs with a reference model.

3. Results:

The testbench checks both read outputs against expected values and displays PASS/FAIL messages.

4. Issues Faced:

Correct handling of register x0 and simultaneous read/write operations is important. The reference model must also update at the correct clock edge.

5. Conclusion:

Successfully designed and verified a 32 × 32 register file with two read ports and one write port.